
The Number Theory Masterclass

Topic 1: Divisibility and Prime Factorization

This is the absolute bedrock of Number Theory. The single most important idea here is the **Fundamental Theorem of Arithmetic**, which states that every integer greater than 1 can be written as a unique product of prime numbers.

Think of prime numbers as the “LEGO blocks” that build all other numbers. Understanding a number’s prime factorization is like understanding its DNA; it reveals its deepest properties.

$$n = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$$

Practice Problem

Find the smallest positive integer n such that $2024n$ is a perfect cube.

(Hint: First, find the prime factorization of 2024. For a number to be a perfect cube, all the exponents in its prime factorization must be multiples of 3. What exponents do you have, and what do they need to become?)

Advanced Challenge

Try: Find all pairs of prime numbers (p, q) such that $p^2 - 2q^2 = 1$.

Topic 2: GCD, LCM, and the Euclidean Algorithm

The **Greatest Common Divisor** and **Least Common Multiple** are fundamental tools. They are linked by a crucial identity:

$$a \times b = \gcd(a, b) \times \text{lcm}(a, b)$$

While you can find the GCD from prime factorizations, the **Euclidean Algorithm** is a brilliant and fast method that works without needing factors. It's based on the principle that $\gcd(a, b) = \gcd(b, a \bmod b)$.

Practice Problem

The sum of two positive integers is 540 and their greatest common divisor is 36. How many such pairs of integers are there?

(Hint: If the GCD of two numbers a and b is g , you can write them as $a = gx$ and $b = gy$, where x and y are coprime. Use this to rephrase the problem.)

Advanced Challenge

Try: Let n be a positive integer. Find the greatest common divisor of $n^2 + 1$ and $(n + 1)^2 + 1$.

Topic 3: Modular Arithmetic (The Language of Remainders)

This is perhaps the most powerful tool in Olympiad Number Theory. We write $a \equiv b \pmod{m}$, which means “ a is congruent to b modulo m ”. This is the same as saying $m \mid (a - b)$.

Think of it as “[clock arithmetic](#).” You can add, subtract, and multiply these congruences, which makes it perfect for simplifying problems with large numbers or proving divisibility rules.

$$a \equiv b \pmod{m} \iff m \mid (a - b)$$

Here are the fundamental rules of modular arithmetic:

Addition: If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then $a + c \equiv b + d \pmod{m}$.

Subtraction: If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then $a - c \equiv b - d \pmod{m}$.

Multiplication: If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then $ac \equiv bd \pmod{m}$.

Exponentiation: If $a \equiv b \pmod{m}$, then $a^k \equiv b^k \pmod{m}$ for any positive integer k .

Practice Problem

Find the last digit of the number 7^{2025} .

(Hint: Finding the last digit is the same as finding the remainder when divided by 10. Look for a pattern in the powers of 7 modulo 10. The pattern will be short and will repeat.)

Advanced Challenge

Try: Prove that for any non-negative integer n , the number $2^{2^{2n+1}} + 3$ is divisible by 7.

Topic 4: Linear Diophantine Equations

These are algebraic equations where we only seek integer solutions. The most common form is $ax + by = c$.

A key theorem, known as **Bezout's Identity**, states that this equation has integer solutions for x and y if and only if $\gcd(a, b)$ divides c . Once you find one solution (x_0, y_0) , you can find all other integer solutions.

$$ax + by = c \text{ has integer solutions} \iff \gcd(a, b) \mid c$$

Practice Problem

A student buys some pencils for 15 Taka each and some erasers for 7 Taka each. If she spends exactly 121 Taka in total, what is the maximum number of items she could have bought?

(Hint: Set up the equation $15p + 7e = 121$, where $p, e \in \mathbb{Z}^+$. Find one solution, then find the general form for all solutions to maximize $p + e$.)

Advanced Challenge

Try: A frog is at position 0 on an infinite number line. It can jump 15 units to the right or 21 units to the left. What is the closest positive integer position to 0 that the frog can reach?

Topic 5: Number of Divisors and Sum of Divisors

For a number with prime factorization $n = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$, we have powerful formulas:

Number of divisors: $d(n) = (a_1 + 1)(a_2 + 1) \cdots (a_k + 1)$

Sum of divisors: $\sigma(n) = \prod_{i=1}^k \frac{p_i^{a_i+1} - 1}{p_i - 1}$

$$d(n) = \prod_{i=1}^k (a_i + 1) \quad \text{and} \quad \sigma(n) = \prod_{i=1}^k \frac{p_i^{a_i+1} - 1}{p_i - 1}$$

A key property of these functions is that they are **multiplicative**. This means that if $\gcd(a, b) = 1$, then $d(ab) = d(a)d(b)$ and $\sigma(ab) = \sigma(a)\sigma(b)$. This is extremely useful for calculating these values for large numbers by breaking them down into their prime power components.

Practice Problem

Find the smallest positive integer that has exactly 15 divisors.

(Hint: You need exponents such that $(a_1 + 1)(a_2 + 1) \cdots = 15$. Find ways to write 15 as a product. To make the number smallest, use the smallest prime bases for the largest exponents.)

Advanced Challenge

Try: A positive integer n is called a **perfect number** if the sum of its positive divisors, $\sigma(n)$, is equal to $2n$. Prove that if $2^k - 1$ is a prime number (known as a Mersenne prime), then the number $n = 2^{k-1}(2^k - 1)$ is a perfect number.

σ (sigma) is the symbol for the sum of divisors function.

$\prod$ (capital pi) denotes a product, analogous to $\sum$ (capital sigma) for sums.

Topic 6: Fermat's Little Theorem (FLT)

A fantastic shortcut for problems involving large powers and prime moduli. This is one of the most elegant results in number theory.

Fermat's Little Theorem: If p is a prime number, then for any integer a such that $p \nmid a$, we have $a^{p-1} \equiv 1 \pmod{p}$.

$$a^{p-1} \equiv 1 \pmod{p} \quad (\text{when } p \text{ is prime and } p \nmid a)$$

$$a^p \equiv a \pmod{p} \quad (\text{always works when } p \text{ is prime})$$

Practice Problem

Find the remainder when 2^{1000} is divided by 13.

(Hint: 13 is prime. By FLT, $2^{12} \equiv 1 \pmod{13}$. Simplify the exponent 1000 by considering its remainder when divided by 12.)

Advanced Challenge

Try: Let p be a prime number. Show that $1^p + 2^p + \cdots + (p-1)^p$ is divisible by p .

Topic 7: Base Systems

Numbers can be written in any integer base $b \geq 2$. A number written as $(d_k d_{k-1} \dots d_1 d_0)_b$ equals $d_k b^k + d_{k-1} b^{k-1} + \dots + d_1 b^1 + d_0 b^0$ in base-10.

Understanding different bases can provide novel ways to solve problems and reveal hidden patterns.

$$(d_k d_{k-1} \dots d_1 d_0)_b = \sum_{i=0}^k d_i b^i$$

Practice Problem

A number is written as $(abab)_c$ in base c (where a and b are digits in that base, with $a \neq 0$). Show that this number is divisible by $c^2 + 1$.

(Hint: Convert to base-10: $a \cdot c^3 + b \cdot c^2 + a \cdot c + b$. Try to factor this algebraic expression.)

Advanced Challenge

Try: A number N is written as $\mathbf{abc}$ in base 7 and as $\mathbf{cba}$ in base 9. Find the value of N in base 10.

Topic 8: Properties of Perfect Squares

Knowing properties of squares can quickly prove that some equations have no integer solutions. These are powerful tools for [proof by contradiction](#).

Last digits: The last digit of a square must be in $\{0, 1, 4, 5, 6, 9\}$

Modulo 4: $n^2 \equiv 0$ or $1 \pmod{4}$ (never 2 or 3)

Modulo 3: $n^2 \equiv 0$ or $1 \pmod{3}$ (never 2)

Practice Problem

Prove that the sum of the squares of three consecutive integers can never be a perfect square.

(Hint: Let the integers be $n-1, n, n+1$. Their sum of squares is $(n-1)^2 + n^2 + (n+1)^2 = 3n^2 + 2$. Analyze this expression modulo 3.)

Advanced Challenge

Try: Prove that the sum of the squares of any five consecutive integers can never be a perfect square.

Topic 9: Euler's Totient Theorem

This generalizes Fermat's Little Theorem to work with any modulus, not just primes. The **totient function** $\varphi(n)$ counts the positive integers up to n that are relatively prime to n .

Euler's Totient Theorem: If $\gcd(a, n) = 1$, then $a^{\varphi(n)} \equiv 1 \pmod{n}$.

$$\text{Totient function: } \varphi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right)$$

$$\text{Totient Theorem: } a^{\varphi(n)} \equiv 1 \pmod{n} \quad (\text{when } \gcd(a, n) = 1)$$

Practice Problem

Find the last two digits of 3^{2024} .

(Hint: This is equivalent to finding $3^{2024} \pmod{100}$. Calculate $\varphi(100) = \varphi(2^2 \cdot 5^2) = 100(1 - 1/2)(1 - 1/5) = 40$. Use this to simplify the exponent.)

Advanced Challenge

Try: Find the last three digits of 7^{9999} .

φ (phi) is the symbol for the totient function.

$\prod$ (capital pi) denotes a product, analogous to $\sum$ (capital sigma) for sums.

Topic 10: Chinese Remainder Theorem (CRT)

While most theorems so far help analyze or simplify existing numbers, the CRT is a master tool of **synthesis**. It allows us to construct a number that satisfies multiple, seemingly unrelated remainder conditions simultaneously. It's the mathematical equivalent of picking a lock for which you have several different keys.

The theorem states that if you have a system of congruences where the moduli are pairwise coprime, a unique solution exists modulo their least common multiple (LCM). For pairwise coprime numbers, the LCM is simply their product, $N = n_1 n_2 \cdots n_k$. This is a powerful idea: it means that in any block of N consecutive integers, there is *exactly one* number that satisfies all the conditions. Once you find this single “base” solution, all other solutions are found simply by adding or subtracting multiples of N .

$$\begin{cases} x \equiv a_1 \pmod{n_1} \\ x \equiv a_2 \pmod{n_2} \\ \vdots \\ x \equiv a_k \pmod{n_k} \end{cases} \quad \text{has a unique solution mod } N = n_1 n_2 \cdots n_k$$

A Walkthrough Example: Iterative Substitution

Let's solve a classic CRT problem. We want to find the smallest positive integer x such that: $x \equiv 2 \pmod{3}$, $x \equiv 3 \pmod{5}$, and $x \equiv 2 \pmod{7}$.

Step 1: From $x \equiv 2 \pmod{3}$, we know $x = 3k + 2$.

Step 2: Substitute this into the second congruence:

$$\begin{aligned} 3k + 2 &\equiv 3 \pmod{5} \\ 3k &\equiv 1 \pmod{5} \\ k &\equiv 2 \pmod{5} \quad (\text{multiplying by } 2) \end{aligned}$$

So, $k = 5j + 2$. We update x : $x = 3(5j + 2) + 2 = 15j + 8$.

Step 3: Substitute this new form of x into the third congruence:

$$\begin{aligned} 15j + 8 &\equiv 2 \pmod{7} \\ j + 1 &\equiv 2 \pmod{7} \quad (\text{since } 15 \equiv 1, 8 \equiv 1) \\ j &\equiv 1 \pmod{7} \end{aligned}$$

So, $j = 7m + 1$.

Step 4: Substitute back to find the general solution for x :

$$\begin{aligned} x &= 15j + 8 \\ x &= 15(7m + 1) + 8 \\ x &= 105m + 15 + 8 \\ x &= 105m + 23 \end{aligned}$$

The smallest positive integer solution is when $m = 0$, which gives $\mathbf{x} = \mathbf{23}$.

An Alternative Method: Direct Construction

This method is more systematic. It constructs the solution from pieces.

Step 1: Define moduli $n_1 = 3, n_2 = 5, n_3 = 7$ and their product $N = 105$.

Step 2: Calculate partial products $N_i = N/n_i$. $N_1 = 105/3 = 35$, $N_2 = 105/5 = 21$, $N_3 = 105/7 = 15$.

Step 3: Find the modular inverse y_i for each N_i such that $N_i y_i \equiv 1 \pmod{n_i}$.

- $35y_1 \equiv 1 \pmod{3} \implies 2y_1 \equiv 1 \pmod{3} \implies \mathbf{y_1 = 2}$.
- $21y_2 \equiv 1 \pmod{5} \implies 1y_2 \equiv 1 \pmod{5} \implies \mathbf{y_2 = 1}$.
- $15y_3 \equiv 1 \pmod{7} \implies 1y_3 \equiv 1 \pmod{7} \implies \mathbf{y_3 = 1}$.

Step 4: Construct the solution using the formula $x \equiv \sum a_i N_i y_i \pmod{N}$.

$$\begin{aligned}x &\equiv a_1 N_1 y_1 + a_2 N_2 y_2 + a_3 N_3 y_3 \pmod{105} \\x &\equiv (2)(35)(2) + (3)(21)(1) + (2)(15)(1) \pmod{105} \\x &\equiv 140 + 63 + 30 \pmod{105} \\x &\equiv 233 \pmod{105} \\x &\equiv \mathbf{23} \pmod{105}\end{aligned}$$

This gives the same smallest positive solution, 23.

What If the Moduli Are Not Coprime?

The standard CRT requires pairwise coprime moduli, but what if they are not? Consider a system of two congruences:

$$\begin{cases} x \equiv a \pmod{m} \\ x \equiv b \pmod{n} \end{cases}$$

From the first congruence, $x = mk + a$ for some integer k . Substituting this into the second gives $mk + a \equiv b \pmod{n}$, or $mk \equiv b - a \pmod{n}$.

This is a linear congruence in the variable k . As we know from the topic on Linear Diophantine Equations, this has a solution for k if and only if $\gcd(m, n)$ divides $b - a$. This gives us the crucial condition for a solution to exist:

$$\text{A solution exists} \iff a \equiv b \pmod{\gcd(m, n)}$$

If this condition holds, we can solve for k and substitute back to find x . The solution will be unique modulo $\text{lcm}(m, n)$. This method allows us to solve systems of congruences one pair at a time, even when the moduli are not coprime.

Practice Problem

Find the smallest positive integer that leaves a remainder of 1 when divided by 4, a remainder of 4 when divided by 5, and is divisible by 7.

(Hint: The conditions are $x \equiv 1 \pmod{4}$, $x \equiv 4 \pmod{5}$, and $x \equiv 0 \pmod{7}$. Use the substitution method from the walkthrough.)

Advanced Challenge

Try: A band of 15 pirates wants to share a treasure of gold coins. When they try to divide the coins into 15 equal piles, they find that 7 coins are left over. They get into a fight over the extra coins, and 3 pirates are thrown overboard. The remaining 12 pirates try to divide the coins again, but this time 1 coin is left over. What is the smallest possible number of coins in the treasure?

Topic 11: Wilson's Theorem

An elegant criterion for primality that connects factorials with modular arithmetic. This is both beautiful and practically useful.

Wilson's Theorem: An integer $p > 1$ is a prime number if and only if $(p-1)! \equiv -1 \pmod{p}$.

$$p \text{ is prime} \iff (p-1)! \equiv -1 \pmod{p}$$

This is a very sharp tool in theoretical problems involving factorials and prime moduli.

Practice Problem

Find the remainder when $16!$ is divided by 19.

(Hint: Start with Wilson's Theorem for $p = 19$: $18! \equiv -1 \pmod{19}$. Rewrite $18! = 18 \times 17 \times 16!$. Replace 18 and 17 with their simpler equivalents modulo 19 and solve for $16!$.)

Advanced Challenge

Try: Let p be a prime number. Find the value of the sum $S = 1! + 2! + 3! + \cdots + (p-1)!$ modulo p .

Topic 12: Legendre's Formula

This formula gives the exact exponent of a prime p in the prime factorization of $n!$. It's essential for problems involving factorials and divisibility.

Legendre's Formula: The exponent of prime p in $n!$, denoted $E_p(n!)$, is:

$$E_p(n!) = \sum_{k=1}^{\infty} \left\lfloor \frac{n}{p^k} \right\rfloor = \left\lfloor \frac{n}{p} \right\rfloor + \left\lfloor \frac{n}{p^2} \right\rfloor + \left\lfloor \frac{n}{p^3} \right\rfloor + \cdots$$

The sum is finite, as the terms become zero when $p^k > n$.

Practice Problem

How many trailing zeros does the number $1000!$ end with?

(Hint: A trailing zero comes from a factor of $10 = 2 \times 5$. The number of zeros is limited by the smaller of the exponents of 2 and 5 in $1000!$. Use Legendre's formula to find $E_5(1000!)$ and $E_2(1000!)$.)

Advanced Challenge

Try: What is the last non-zero digit of $20!$?